Time Series Forecasting: Seasonal Regression with CatBoost Residuals

1. Introduction

The objective of this project is to develop an accurate time series forecasting model for hourly wind energy generation. The model is evaluated against a Seasonal Naive benchmark using point forecast accuracy and prediction interval quality.

The forecasting task requires predicting the next 48 hours of hourly observations. The final model must produce both point forecasts and 95% prediction intervals.

The main objectives are:

- Improve point forecast accuracy compared with the Seasonal Naive benchmark.
- Reduce forecasting errors measured by RMSE and MAPE.
- Construct reliable 95% prediction intervals.
- Evaluate the model using rolling validation.
- Produce a reproducible forecasting pipeline.

The final model is a Seasonal Regression model enhanced with CatBoost residual corrections.

2. Data and Problem Formulation

The dataset contains historical hourly observations of wind energy generation and weather-related variables.

The target variable is:

- `y` — hourly wind energy generation.

The weather variables used as predictors are:

- `cloud_cover`
- `temperature`
- `wind_speed`
- `pressure`
- `humidity`
- `daylight`
- `air_density`

The dataset contains historical observations in `train.csv`, while `weather_forecast.csv` provides weather information for the future forecasting horizon.

The forecasting task is formulated as:

Forecast = Seasonal Baseline + Predicted Residual

More specifically:

$\hat{y}(t) = y(t - 24) + \hat{r}(t)$

where:

- $y(t - 24)$ is the seasonal baseline;
- $\hat{r}(t)$ is the predicted residual;
- $\hat{y}(t)$ is the final forecast.

This formulation allows the model to learn deviations from the daily seasonal pattern instead of predicting the complete target value directly.

3. Methodology

3.1. Seasonal Baseline

The model uses a daily seasonal baseline.

For each hour, the initial forecast is the observed value from the same hour of the previous day:

$\hat{y}_{baseline}(t) = y(t - 24)$

This approach captures the daily periodicity of the series.

However, hourly wind energy generation can vary considerably due to weather conditions. Therefore, the seasonal baseline is corrected using a machine learning model.

3.2. Feature Engineering

The model uses calendar features, cyclical transformations, historical observations, and weather variables.

Calendar features

The following features are extracted from the timestamp:

- Hour of the day
- Day of the week
- Day of the year
- Month

These variables help capture temporal patterns and seasonal changes.

Cyclical encoding

Hour and day of the year are represented using sine and cosine transformations.

For the hour of the day:

$hour_sin = \sin(2\pi \times hour / 24)$

$hour_cos = \cos(2\pi \times hour / 24)$

Similar transformations are applied to the day of the year.

Cyclical encoding preserves the circular nature of time. For example, hours 23 and 0 are treated as temporally adjacent.

Historical features

The model incorporates:

- Seasonal baseline: $y(t - 24)$
- Lag 48: $y(t - 48)$
- Lag 168: $y(t - 168)$
- Change over the previous day:

$change_24 = y(t - 24) - y(t - 48)$

These features provide information about recent generation levels and weekly seasonality.

Weather features

Weather variables are included to capture the relationship between meteorological conditions and wind energy generation.

3.3. CatBoost Residual Model

Instead of predicting the target directly, CatBoost predicts the residual between the actual observation and the seasonal baseline:

$r(t) = y(t) - y(t - 24)$

The final prediction is calculated as:

$\hat{y}(t) = y(t - 24) + \hat{r}(t)$

The CatBoost model uses the following configuration:

Parameter	Value
Iterations	600
Depth	5
Learning rate	0.03
Loss function	RMSE
L2 leaf regularization	5
Random seed	42

CatBoost was selected because it can model nonlinear relationships between weather conditions, temporal features, and wind energy generation.

3.4. Recursive Forecasting

The forecasting procedure generates predictions sequentially.

For each future hour:

1. The daily seasonal baseline is calculated.
2. Calendar and weather features are constructed.
3. CatBoost predicts the residual.
4. The residual is added to the seasonal baseline.
5. The forecast is constrained to be non-negative.
6. The predicted value is stored for subsequent forecasts.

This procedure allows the model to generate forecasts for the complete 48-hour horizon.

4. Evaluation Metrics

The model is evaluated using three metrics.

4.1. Root Mean Squared Error (RMSE)

RMSE measures the magnitude of forecasting errors and penalizes large errors more heavily.

$$RMSE = \sqrt{[\Sigma(y(t) - \hat{y}(t))^2 / n]}$$

Lower RMSE indicates better forecasting accuracy.

4.2. Mean Absolute Percentage Error (MAPE)

MAPE expresses the forecasting error as a percentage of the actual value.

$$MAPE = (100\% / n) \times \Sigma |(y(t) - \hat{y}(t)) / y(t)|$$

Lower MAPE indicates better relative forecasting accuracy.

MAPE can become unstable when actual observations are close to zero. Therefore, it should be interpreted together with RMSE.

4.3. Winkler Score

The Winkler score evaluates the quality of prediction intervals.

For a nominal 95% prediction interval:

$$W(t) = (U(t) - L(t)) + (2 / \alpha) \times (L(t) - y(t))_+ + (2 / \alpha) \times (y(t) - U(t))_+$$

where:

- $L(t)$ is the lower interval bound;
- $U(t)$ is the upper interval bound;
- $\alpha = 0.05$;
- $(x)_+ = \max(x, 0)$.

The final score is:

$$Winkler = \Sigma W(t) / n$$

A lower Winkler score indicates a better prediction interval.

The score penalizes both excessively wide intervals and observations falling outside the interval.

4.4. Coverage

Coverage measures the proportion of actual observations contained within the prediction intervals.

$$Coverage = (1 / n) \times \Sigma I(L(t) \leq y(t) \leq U(t))$$

For a well-calibrated 95% prediction interval, coverage should be close to 0.95.

However, high coverage alone does not guarantee good interval quality because excessively wide intervals can contain almost all observations.

5. Validation Strategy

The model is evaluated using rolling validation with a 48-hour forecasting horizon.

The validation offsets are:

168, 336, and 672 observations.

These offsets correspond to one, two, and four weeks of historical observations.

For each validation window:

1. The model is fitted using only the historical observations available before the validation period.
2. The next 48 hours are forecast.
3. Predictions are compared with actual observations.
4. RMSE and MAPE are calculated.

This approach evaluates the model under conditions similar to the actual forecasting task.

It also helps identify whether model performance changes across different historical periods.

6. Prediction Interval Calibration

The model generates point forecasts first. Prediction intervals are then constructed using empirical residual quantiles.

The validation errors are defined as:

$e(t) = y(t) - \hat{y}(t)$

The lower and upper error quantiles are estimated as:

Lower quantile = Quantile(e, 0.025)

Upper quantile = Quantile(e, 0.975)

The prediction interval is calculated as:

Lower bound = max(0, Forecast + Lower quantile)

Upper bound = max(Lower bound, Forecast + Upper quantile)

The lower bound is constrained to be non-negative because wind energy generation cannot be negative.

The interval calibration is based on empirical forecasting errors rather than assuming normally distributed residuals.

7. Results

7.1. Point Forecast Comparison

The following results were obtained on the 48-hour evaluation window.

Model	RMSE	MAPE
Seasonal Naive	120.404	51.088
Seasonal Regression + CatBoost	102.524	39.629

The proposed model improves both point forecasting metrics.

The relative RMSE improvement is:

RMSE Improvement = ((120.404 − 102.524) / 120.404) × 100% ≈ 14.85%

The relative MAPE improvement is:

MAPE Improvement = ((51.088 − 39.629) / 51.088) × 100% ≈ 22.43%

These results indicate that incorporating weather information and nonlinear residual corrections improves the point forecast compared with the Seasonal Naive benchmark.

The model achieves approximately 14.85% lower RMSE and 22.43% lower MAPE, demonstrating improved forecasting accuracy.

7.2. Prediction Interval Comparison

The prediction interval results are:

Model	Winkler	Coverage
Seasonal Naive	568.464	1.000
Seasonal Regression + CatBoost	768.133	1.000

Both models achieve complete coverage on the evaluated 48-hour window.

However, the proposed model has a higher Winkler score.

This indicates that although the model improves point forecasting accuracy, its prediction intervals are less efficient than those of the Seasonal Naive benchmark.

The main reason is the width of the prediction intervals. The empirical residual quantiles produce relatively wide intervals, which increase the Winkler score even when all actual observations are covered.

Therefore, prediction interval calibration remains an important area for further improvement.

7.3. Rolling Validation Results

The rolling validation results for the model are presented below.

Validation window	RMSE	MAPE
168	194.754	118.403
336	298.788	100.690
672	218.736	36.735

The results show that forecasting accuracy varies across historical periods.

This variation suggests that the model's performance depends on the conditions present in each validation window.

In particular, the model may have difficulty predicting periods with substantial changes in wind energy generation.

The rolling validation results also indicate that a single validation window may not fully represent the model's expected performance.

8. Discussion

The results demonstrate that the Seasonal Regression + CatBoost Residuals model can improve point forecasting accuracy compared with the Seasonal Naive benchmark.

The main advantages of the model are:

- It captures daily seasonality through the seasonal baseline.
- It incorporates weather information.
- It models nonlinear relationships using CatBoost.
- It uses historical lags to capture recent generation patterns.
- It produces non-negative forecasts.
- It generates prediction intervals using empirical residual quantiles.

However, several limitations remain.

First, the model uses a recursive forecasting procedure. Errors made in earlier forecast steps can influence subsequent predictions.

Second, the residual model assumes that the relationship between the seasonal baseline, weather conditions, and generation remains sufficiently stable over the forecasting horizon.

Third, the prediction intervals are relatively wide. Although coverage is high, the Winkler score indicates that the intervals could be more efficient.

Finally, the rolling validation results show considerable variation in performance across different historical periods. This suggests that additional validation windows and further model calibration may be beneficial.

9. Conclusion

This project developed a forecasting model based on Seasonal Regression and CatBoost residual correction.

The model combines a daily seasonal baseline with calendar features, historical observations, and weather variables.

The evaluation results show that the proposed model improves point forecasting accuracy compared with the Seasonal Naive benchmark.

The RMSE decreases from 120.404 to 102.524, while MAPE decreases from 51.088 to 39.629.

The relative RMSE improvement is approximately 14.85%, and the relative MAPE improvement is approximately 22.43%.

However, the Winkler score increases fro